Advanced information, computation, communication I
EPFL - Fall semester 2026-2027

Week 2 — solutions

September 18, 2026

Exercise 1. *Which of the following represents the CNF and DNF forms of the compound proposition:*

$$(p \vee \neg q) \wedge (\neg p \vee q)$$

- ◯ CNF: $(p \vee \neg q) \wedge (\neg p \vee q)$, DNF: $(p \vee \neg q) \wedge (\neg p \vee q)$
- ✓ CNF: $(p \vee \neg q) \wedge (\neg p \vee q)$, DNF: $(p \wedge q) \vee (\neg p \wedge \neg q)$
- ◯ CNF: $(p \vee q) \wedge (\neg p \vee \neg q)$, DNF: $(p \wedge q) \vee (\neg p \wedge \neg q)$
- ◯ CNF: $(p \wedge q) \vee (\neg p \wedge \neg q)$, DNF: $(p \vee q) \wedge (\neg p \vee \neg q)$

Given the proposition $(p \vee \neg q) \wedge (\neg p \vee q)$:

- It is already in CNF as it's a conjunction of two clauses.
- For DNF, we first apply the distributive law and get:

$$\begin{aligned} &(p \vee \neg q) \wedge (\neg p \vee q) \\ \equiv &[(p \vee \neg q) \wedge (\neg p)] \vee [(p \vee \neg q) \wedge (q)] \end{aligned}$$

We apply the distributive law again and get:

$$\begin{aligned} &[(p \vee \neg q) \wedge (\neg p)] \vee [(p \vee \neg q) \wedge (q)] \\ \equiv &\left[\underbrace{(p \wedge \neg p)}_{\equiv F} \vee (\neg q \wedge \neg p)\right] \vee \left[(p \wedge q) \vee \underbrace{(\neg q \wedge q)}_{\equiv F}\right] \\ \equiv &(\neg q \wedge \neg p) \vee (p \wedge q) \end{aligned}$$

This gives the CNF as $(p \vee \neg q) \wedge (\neg p \vee q)$ and the DNF as $(p \wedge q) \vee (\neg p \wedge \neg q)$. Thus, the correct option is the second one.

Exercise 2.

1. *Find a DNF of $p \oplus (q \wedge r)$ without using a truth table.*
   Recall that $a \oplus b \equiv (a \wedge \neg b) \vee (\neg a \wedge b)$.
   Using this, the given proposition $p \oplus (q \wedge r)$ can be rewritten as:

$$\begin{aligned} &(p \wedge \neg(q \wedge r)) \vee (\neg p \wedge (q \wedge r)) \\ \equiv &(p \wedge (\neg q \vee \neg r)) \vee (\neg p \wedge (q \wedge r)) \text{ De Morgan's Law} \\ \equiv &((p \wedge \neg q) \vee (p \wedge \neg r)) \vee (\neg p \wedge (q \wedge r)) \text{ Distributive Law} \\ \equiv &(p \wedge \neg q) \vee (p \wedge \neg r) \vee (\neg p \wedge q \wedge r) \end{aligned}$$

This is a disjunction of conjunctions, which means it's in DNF.

$$\text{DNF: } (p \wedge \neg q) \vee (p \wedge \neg r) \vee (\neg p \wedge q \wedge r)$$

2. *Find a CNF of* $[(p \to q) \vee (r \to s)] \wedge u$ *without using a truth table.*
   Firstly, we rewrite $(p \to q) \vee (r \to s)$ using $a \to b \equiv \neg a \vee b$:

   $$(\neg p \vee q) \vee (\neg r \vee s)$$

   Which is equivalent to:

   $$\neg p \vee q \vee \neg r \vee s$$

   By replacing the in original expression, we get

   $$[\neg p \vee q \vee \neg r \vee s] \wedge u$$

   Which is a CNF.

Exercise 3. (Warm-up; unlike the multiple choice questions at the final exam, more than a single answer may be correct.). *The law regarding alcohol drinking in Switzerland states the following: "If you are drinking strong alcohol, then you must be at least 18 years old." (for all* x, $A(x) \to 18(x)$*). Assume that you are hired as a police officer at Satellite and need to enforce the above law. Which student(s) are you going to arrest and why:*

- ◯ *The 17-year old student drinking soda.*
- ✓ *The 17-year old student drinking vodka.*
- ◯ *The 18-year old student drinking water.*
- ◯ *The 18-year old student drinking tequila.*

Next Tuesday, Satellite organizes a cabaret night open to all students under the following conditions: "Everyone must be at least 18 years old and drink strong alcohol" (for all x, $A(x) \wedge 18(x)$*). As previously, which student(s) are you going to arrest and why:*

- ✓ *The 17-year old student drinking soda.*
- ✓ *The 17-year old student drinking vodka.*
- ✓ *The 18-year old student drinking water.*
- ◯ *The 18-year old student drinking tequila.*

Exercise 4. *(Rosen, exercise 8, 1.4.14 in* 8th *edition) Translate these statements into English, where* $R(x)$ *is "*x *is a rabbit" and* $H(x)$ *is "*x *hops" and the domain consists of all animals.*

1. $\forall x(R(x) \to H(x))$: All rabbits hop.
2. $\exists x(R(x) \to H(x))$: Multiple possible solutions (they are all correct):
   - There is an animal that hops if it's a rabbit
   - There exists an animal where if it is a rabbit, then it hops
   - There exists an animal such that if it is a rabbit, then it hops

3. $\forall x(R(x) \wedge H(x))$: All animals are rabbits and also hop.
4. $\exists x(R(x) \wedge H(x))$: There is a rabbit that also hops.

Exercise 5.

1. *Which expressions below are equivalent to* $\neg(\forall x \exists y\, P(x,y))$*. Explain.*

   ✓ $\exists x \forall y\ \neg P(x,y)$;
   ◯ $\exists x \exists y\ \neg P(x,y)$.

   The formula
   $$\forall x \exists y\, P(x,y)$$
   states that for every x, there exists some y such that $P(x,y)$ is true.
   We find its negation by moving the negation inward step by step:
   $$\neg(\forall x \exists y\, P(x,y)) \leftrightarrow \exists x \neg(\exists y\, P(x,y)) \leftrightarrow \exists x \forall y\, \neg P(x,y).$$
   In words, the negation says that there exists some x for which no y makes $P(x,y)$ true, i.e., $P(x,y)$ is false for every y.
   Therefore, the first expression is equivalent. The second is not, since it only requires $P(x,y)$ to fail for one pair (x,y).

Exercise 6. *(Rosen, exercise 9, 1.5.8 in* 8th *edition) Let* $L(x,y)$ *be the statement "x loves y", where the domain for both x and y consists of all people in the world. Use quantifiers to express each of these statements:*

1. *Everybody loves Sharon.*

   $\forall x\ L(x, \text{Sharon})$.

2. *Everybody loves somebody.*

   $\forall x \exists y\ L(x,y)$.

3. *There is somebody whom everybody loves.*

   $\exists x \forall y\ L(y,x)$.

4. *Nobody loves everybody.*

   $\forall x \exists y\ \neg L(x,y)$.

5. *There is somebody whom Daisy does not love.*

   $\exists x\ \neg L(\text{Daisy}, x)$.

6. *There is somebody whom no one loves.*

   $\exists x \forall y \neg L(y,x)$.

7. *There is exactly one person whom everybody loves.*

   $\exists x\ (\forall y\ L(y,x) \wedge \forall z((\forall w\ L(w,z)) \rightarrow z = x))$.

8. *There are exactly two people whom Marsellus loves.*

   $\exists x \exists y\ (x \neq y \wedge L(\text{Marsellus}, x) \wedge L(\text{Marsellus}, y) \wedge \forall z(L(\text{Marsellus}, z) \rightarrow (z = x \vee z = y)))$.

9. *Everyone loves himself or herself.*

   $\forall x\ L(x,x)$.

10. *There is someone who loves no one besides himself or herself.*

    There are two ways to interpret this question:

    A) To interpret it as "there is someone who doesn't love anyone who is not himself/herself, but may or may not love himself/herself," in which case, the solution is:

    $\exists x \forall y\ (L(x,y) \to x = y)$.

    B) To interpret it as "there is someone who doesn't love anyone who is not himself/herself, but loves himself/herself," in which case, the solution is:

    $\exists x \forall y\ (L(x,x) \wedge (L(x,y) \to x = y))$.

Exercise 7. *Given the two statements below, where the domain of discourse is* $\mathbf{R}$ *for both* x *and* y*,*

$$\exists y \forall x (x \neq 0 \to xy = 1) \qquad\qquad \exists x \forall y (xy < 0 \to xy > 0)$$

◯ *They are both false.*

◯ *Only the first is true.*

✓ *Only the second is true.*

◯ *They are both true.*

The first statement is "obviously incorrect" because not all non-zero x-values can be the inverse of some particular y-value. But let's do it more carefully: if the first statement is False, then its negation must be True. So, consider the negation of the first statement:

$$\begin{aligned} \neg\big(\exists y \forall x (x \neq 0 \to xy = 1)\big) &\equiv \neg\big(\exists y \forall x (\neg(x \neq 0) \vee xy = 1)\big) \\ &\equiv \forall y \exists x (x \neq 0 \wedge \neg(xy = 1)) \\ &\equiv \forall y \exists x (x \neq 0 \wedge xy \neq 1). \end{aligned}$$

We have to show that this final statement "$\forall y \exists x (x \neq 0 \wedge xy \neq 1)$" is True: for $y = 0$ one can take any x with $x \neq 0$ (because $xy = 0$ and thus $xy \neq 1$), and for $y \neq 0$ one can take $x = \frac{2}{y}$ (because then $x \neq 0$ and $xy = 2$ and thus $xy \neq 1$). So, irrespective of the value of y, a non-zero x-value can be found such that xy is not equal to one, proving that indeed "$\forall y \exists x (x \neq 0 \wedge xy \neq 1)$" is True. Because the negation of the first statement it True, the first statement is False.

For the statement "$\exists x \forall y (xy < 0 \to xy > 0)$", consider $x = 0$, then for all y it is the case that $xy < 0$ is False and thus "$xy < 0 \to xy > 0$" is True. It follows that an x-value exists such that for all y-values the statement "$xy < 0 \to xy > 0$" is True: thus the second statement is True.

Exercise 8.

Consider the two statements below, where $P(x,y)$ *is a propositional function and the domain of discourse is* $\mathbf{Z}_{\geq 0}$ *for* x*,* y *and* z*:*

$$(\exists y \forall x\, P(x,y)) \to (\forall x \exists y\, P(x,y)) \qquad\qquad (\neg\exists x\ x^x = x!) \to \forall y, z\ y \neq z.$$

◯ *They are both false.*

◯ *Only the first is true.*

◯ *Only the second is true.*

✓ *They are both true.*

If $\exists y \forall x\, P(x,y)$ is True, there is some value $\tilde{y}$ (formally: use existential instantiation) such that $\forall x P(x,\tilde{y})$ is True. Thus (formally using existential generalization) $\forall x \exists y\, P(x,y)$. It follows that the first statement is True.

The statement $\neg\exists x\; x^x = x!$ is equivalent to $\forall x\; x^x \neq x!$. Because $x = 1$ is in the domain $\mathbf{Z}_{\geq 0}$ for x it follows from $\forall x\; x^x \neq x!$ (formally: use universal instantiation) that $1^1 \neq 1!$; but $1^1 = 1 = 1!$ so $\forall x\; x^x \neq x!$ is False. Because the statement "False$\rightarrow q$" is True for any (logical) value q, it follows that the second statement is True.

Exercise 9. *Let E be a set of endpoints on a network, let P be a set of paths connecting those endpoints, and let $C(p,x,y)$ be the proposition that path $p \in P$ connects endpoints x and y with $x, y \in E$.* The statement "there are at least two paths connecting every two distinct endpoints on the network" can be expressed by

◯ $\forall x, y \in E \Big(x \neq y \rightarrow \exists p, q \in P \big(p \neq q \wedge (C(p,x,y) \vee C(q,x,y))\big)\Big)$.

◯ $\forall x, y \in E \Big(x \neq y \wedge \exists p, q \in P \big(p \neq q \wedge C(p,x,y) \wedge C(q,x,y)\big)\Big)$.

✓ $\neg\Big(\exists x, y \in E \big(x \neq y \;\wedge\; \forall p, q \in P \,(p = q \vee \neg C(p,x,y) \vee \neg C(q,x,y))\big)\Big)$.

◯ $\neg\Big(\exists x, y \in E \big(x \neq y \;\wedge\; \forall p, q \in P \,(p = q \wedge \neg C(p,x,y) \wedge \neg C(q,x,y))\big)\Big)$.

Given any two distinct endpoints (i.e., $\forall x, y \in E$ if $x \neq y$ then $\ldots$) there are at least two paths (i.e., $(x \neq y \rightarrow \exists p, q \in P\ (p \neq q \ldots))$ connecting the two distinct endpoints (i.e., $(p \neq q \wedge C(p,x,y) \wedge C(q,x,y))$), leading to the complete logical expression

$$\forall x, y \in E\ \big(x \neq y \rightarrow \exists p, q \in P\ (p \neq q \wedge C(p,x,y) \wedge C(q,x,y))\big).$$

This is equivalent to

$$\neg\neg\Big(\forall x, y \in E\ \big(\neg(x \neq y) \vee \exists p, q \in P\ (p \neq q \wedge C(p,x,y) \wedge C(q,x,y))\big)\Big)$$

and thus to

$$\neg\Big(\exists x, y \in E\ \big(x \neq y \wedge \neg\exists p, q \in P\ (p \neq q \wedge C(p,x,y) \wedge C(q,x,y))\big)\Big)$$

and finally to

$$\neg\Big(\exists x, y \in E \big(x \neq y \;\wedge\; \forall p, q \in P \,(p = q \vee \neg C(p,x,y) \vee \neg C(q,x,y))\big)\Big).$$

The other three possibilities can be seen to be wrong in various different ways.

Exercise 10. *Given the propositional function $T(x)$, the statement $\exists! x\, T(x)$ is logically equivalent to*

✓ $\neg(\forall x\, [T(x) \rightarrow \exists y \neq x\, T(y)])$.

◯ $\exists x \forall y((\neg T(y)) \vee (y = x))$.

◯ $\exists x(T(x) \vee \forall y[(\neg T(y)) \vee (y = x)])$.

◯ $\exists x(T(x) \wedge \forall y[T(y) \wedge (y = x)])$.

The second and third statements are True for the propositional function $T(x)$ with a non-empty domain that is False for all x (and for which the statement $\exists! x\, T(x)$ is thus False). The fourth statement is False for, for instance, the propositional function $T(x)$ that consists of the statement "x equals 1" and where the domain of x contains the element 1 and at least one other element, whereas for that same propositional function the statement $\exists! x\, T(x)$ is True.

This leaves only the first statement, and indeed it is equivalent to $\exists! x\, T(x)$ because $\neg(\forall x\, [T(x) \rightarrow \exists y \neq x\, T(y)])$ is equivalent to $\neg(\forall x\, [\neg T(x) \vee \exists y \neq x\, T(y)])$ and thus equivalent to $\exists x\, T(x) \wedge \forall y \neq x\ \neg T(y)$.

Exercise 11. *Given the propositional functions $G(x)$: "x is a boy", $F(y)$: "y is a girl", and $A(z)$: "z likes computers", the statement "all boys like computers and there is a girl that does not like computers" can be expressed by*

- ✓ $\neg[(\exists y\,(G(y) \wedge \neg A(y))) \vee (\forall x\,(F(x) \rightarrow A(x)))]$.
- ◯ $\neg[(\forall x\,(F(x) \rightarrow A(x))) \vee (\exists y\,(G(y) \rightarrow \neg A(y)))]$.
- ◯ $(\exists x\,(F(x) \rightarrow \neg A(x))) \wedge (\forall y\,(\neg G(y) \vee A(y)))$.
- ◯ $(\forall y\,(G(y) \wedge A(y))) \wedge (\exists x\,(F(x) \wedge \neg A(x)))$.

Using the equivalence $p \rightarrow q \equiv \neg p \vee q$, the first answer is equivalent to $\neg[(\exists y\,(G(y) \wedge \neg A(y))) \vee (\forall x\,(\neg F(x) \vee A(x)))]$, implying it is equivalent to $(\forall y\,(\neg G(y) \vee A(y))) \wedge (\exists x\,(F(x) \wedge \neg A(x)))$ and thus to $(\forall y\,(G(y) \rightarrow A(y))) \wedge (\exists x\,(F(x) \wedge \neg A(x)))$. This says that for all elements of the domain it is the case that if the element is a boy, then that boy like computers ("$\forall y\,(G(y) \rightarrow A(y))$") and that furthermore ("$\wedge$") there exists an element of the domain that is a girl that does not like computers ("$\exists x\,(F(x) \wedge \neg A(x))$"). This corresponds to the statement "all boys like computers and there is a girl that does not like computers". Note that this statement does not imply that there are any boys in the domain – but if there are boys, then those boys like computers.

The other answers are incorrect:

- Twice using the same equivalence $p \rightarrow q \equiv \neg p \vee q$ again, the second answer is equivalent to $\neg[(\forall x\,(\neg F(x) \vee A(x))) \vee (\exists y\,\neg(G(y) \vee \neg A(y)))]$, implying it is equivalent to $(\exists x\,(F(x) \wedge \neg A(x))) \wedge (\forall y\,(G(y) \wedge A(y)))$. This says that there exists an element of the domain that is a girl that does not like computers ("$\exists x\,(F(x) \wedge \neg A(x))$") and that furthermore ("$\wedge$") all elements of the domain are boys that like computers ("$\forall y\,(G(y) \wedge A(y))$"): this is contradictory because the girl (that does not like computers) cannot at the same time be one of the boys (that like computers).
- The third answer says that there exists an element of the domain such that <u>if</u> that element is a girl then that girl does not like computers ("$\exists x\,(F(x) \rightarrow \neg A(x))$") combined with ("$\wedge$") a condition that does not express anything about "girls". It follows that the first part is True if there are no girls in the (non-empty) domain, whereas the original statement says that "there is a girl ...".
- The fourth answer is equivalent to the reformulated version of the second answer (as derived above) because $p \wedge q \equiv q \wedge p$.